

1) k - Thermal conductivity

- The ability of a material to transfer heat energy through conduction

$$k = \frac{W}{m \cdot K} = \frac{kg \cdot m}{s^3 \cdot K} \xrightarrow{\text{in book way}} \frac{m \cdot L}{t^3 \cdot T}$$

α - Thermal diffusivity

$\alpha = \frac{k}{\rho \hat{C}_p}$ → measure the rate of heat transfer inside a material

$$= \frac{m^2}{s} \xrightarrow{\text{in book way}} \frac{L^2}{t}$$

$\hat{C}_p$ - Heat Capacity

- The amount of heat require to raise the temperature of a object by $1^\circ C$ or a unit change in its temperature

$$C_p = \frac{J}{K \cdot kg} = \frac{m^2}{s^2 \cdot K} \xrightarrow{\text{in book way}} \frac{L^2}{t^2 \cdot T}$$

q - Heat flux

$$- q = -k \nabla T$$

- This is the heat flux that is proportional to the temperature gradient

$$q = \frac{W}{t^3}$$

e - Combine energy flux vector

$$e = \left(\underbrace{\frac{1}{2} \rho v^2}_{\text{kinetic}} + \underbrace{\rho \hat{H}}_{\text{energy}} \right) v + \underbrace{[\tau \cdot v]}_{\text{work}} + \underbrace{q}_{\text{heat}}$$

$$e = \frac{W}{t^3}$$

$$2) k_{\text{gas}} < k_{\text{liquid}} < k_{\text{solid}}$$

3) Newton

Fourier

$$\tau_{xy} = -\mu \frac{dv}{dy}$$

$$q_y = -k \frac{dT}{dy}$$

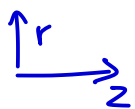
- Transport of momentum
- Involve a vector field
- Tensor
- Doesn't apply for solids

- Transport of heat
- Involve scalar field
- Simple 1D scalar only
- Any liquid, gas, solid

8) $C_p - C_v = R$ is only for gas

For liquids use $C_p - C_v = \frac{\alpha^2 T}{\rho \beta_T}$

9)



$$V_z = \frac{(P_o - P_L) R^2}{4\mu L} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

$\swarrow$
 $V_{z,\text{max}}$

$$V_z = V_{z,\text{max}} \left(1 - \left(\frac{r}{R} \right)^2 \right) \rightarrow V_{z,\text{Ave}} = \frac{1}{\pi R^2} \int_0^{2\pi} \int_0^R V_{z,\text{max}} \left(1 - \left(\frac{r}{R} \right)^2 \right) r dr d\theta$$

$$\text{ke: } \left(\frac{1}{2} \rho V \right) V = \frac{1}{2} \rho V_z^3 \quad V_{z,\text{Ave}} = \frac{V_{z,\text{max}}}{2} \rightarrow 2V_{z,\text{Ave}} = V_{z,\text{max}}$$

$$\Rightarrow KE = \int_0^{2\pi} \int_0^R \frac{1}{2} \rho \left(2V_{z,Ave} \left(1 - \frac{r}{R} \right)^2 \right)^3 r dr d\theta$$

$$= \pi R^2 \int V_{z,Ave}^3$$

$$\frac{1}{2} \rho 8 V_{z,Ave}^3$$

$$4 \rho V_{z,Ave}^3 \cdot \frac{\pi R^2}{4}$$

Put as $V_{z,Ave}$ for easy
to measure as we already
know the flow profile

$$KE = \alpha \left(\frac{1}{2} \dot{m} V_{avg}^2 \right)$$

$$\dot{m} = \int_0^{2\pi} \int_0^R \rho V_{avg} r dr d\theta$$

$$\int_0^{2\pi} \int_0^R \rho \frac{V_{z,max}}{2} r dr d\theta = \rho \frac{V_{z,max}}{2} \pi R^2$$

$$= \pi R^2 \rho \frac{V_{z,max}}{2}$$

$$\Rightarrow KE = \alpha \left(\frac{1}{2} \pi R^2 \rho \frac{V_{z,max}^3}{2} \right)$$

$$= \alpha \left(\frac{1}{4} \pi R^2 \rho V_{z,max}^3 \right)$$

$$\text{Laminar } \alpha = 2 \rightarrow KE = \frac{1}{2} \pi R^2 \rho V_{z,max}^3$$

$$= \pi R^2 \rho V_{z,Ave}^3$$

Same